Similarly, there will be 6 red and 2 white marbles in the box for the third draw if the events A and B have occurred. Hence,

$P(C/A \cap B) = \frac{6}{8}$.

$\therefore \quad P(A \cap B \cap C) = \frac{7}{10} \times \frac{3}{9} \times \frac{6}{8} = \frac{7}{40}$ is the required probability of drawing three marbles in the alternate colours with the first marble being red.

Example 8.19

Three cards are drawn successively from a well-shuffled pack of 52 playing cards. Find the probability all three cards drawn successively is ace without replacing the card after each draw.

Solution:

Let A: all the three cards drawn are aces

At the first draw, there will be 4 aces among 52 cards. Having drawn an ace in the first draw, there will be 3 aces among 51 cards. Similarly, there will be 2 aces among 51 cards for the third draw.

Then, as discussed in Example 8.20, by Theorem 8.5

$$P(A) = \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50}$$

$$P(A) = \frac{1}{5525}.$$

Example 8.20

There are 13 boys and 6 girls in a class. Four students are selected randomly one after another from that class. Find the probability that: (i) all are girls, (ii) first two are boys and next are girls

Solution:

(i) Suppose that

B: all the randomly selected students are girls

There will be 6 girls among 19 students, in total, while selecting the first student; there will be 5 girls among 18 students, in total, while selecting the second student; 4 girls among 17 students, in total, while selecting the third student; and 3 girls among the remaining 16 students, in total, while selecting the fourth student.